

# Late-Layer Geometric Resistance Under In-Context Knowledge Conflict

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July 6, 2026

**Status:** workshop-note draft, small/moderate- $n$  exploratory interpretability result. **Scope:** a narrow, falsification-heavy finding whose central result (§2.6) is a negative one — see §4 before citing.

## Abstract

When a document in context contradicts a language model’s parametric knowledge, does that conflict leave a measurable trace in the residual stream? On isolated, single-sentence synthetic contradictions, yes, robustly: across three instruction-tuned models spanning three architecture families (Qwen3-4B-Instruct, Phi-3-mini, Llama-3.2-3B-Instruct) and three relation types (country→capital, company→founder, book→author), a late-layer ( $\sim 82\text{--}83\%$  depth) hidden state for a same-entity contradiction is geometrically further from a clean baseline than switching to a different entity entirely is (9/9 model×relation combinations agree on direction), while a mid-layer ( $\sim 32\text{--}34\%$  depth) check shows no such gap. We then falsify our own first two explanations for *why* the effect varies in size — first that it tracks per-fact parametric memorization strength (rigidity hypothesis, refuted: Spearman  $\rho = 0.006$ ,  $n = 45$ ), then that it tracks the model successfully outputting the false claim (active-lie hypothesis, refuted: resisting the contradiction costs *more* geometric disruption than complying with it,  $n = 73$  vs. 62). Finally — and this is the central limitation of the whole result — the effect is largely an artifact of the single-sentence framing: scaled to 120 facts across 6 relations with realistic multi-sentence documents (the target fact embedded among other true facts, not isolated), the late-layer gap survives for only 1 of 6 relations and *reverses sign* for the other 5. The mid-layer null is separately reconciled with prior work: a *trained* probe recovers a real but modest signal (AUROC 0.642) where our untrained geometric check found none, consistent with — but quantitatively short of — Zhao et al. (2024)’s reported  $\sim 90\%$  mid-layer probe accuracy.

## 1 Setup

Three relations, 15 facts each (45 total), each posed as Document: {fact}. Q: {question} A: with either the true tail (congruent) or another fact’s real tail substituted (contradictory, a plausible hard negative rather than gibberish). Models confirmed to actually follow in-context overrides before use (checked directly, not assumed, and at the *aggregate* level, not on one example — a single Llama-3 test case looked like a premise failure and was not representative, see §2.2): GPT-2-base and TinyLlama-1.1B-Chat both failed this premise outright at scale and were excluded; Qwen3-4B-Instruct, Phi-3-mini-4k-instruct, and Llama-3.2-3B-Instruct all pass, with real (if uneven) override rates on every relation. Llama-3 required a license acceptance and access token, obtained partway through this work.

## 2 Results

### 2.1 Real-LLM parametric recall (baseline sanity check)

A linear probe on GPT-2 hidden states, trained on 20 countries’ true/false capital sentences, reaches AUROC 0.980 on 10 held-out countries — confirms parametric facts are linearly decodable and generalize to unseen entities, before testing anything harder.

### 2.2 Late-layer decoherence under in-context contradiction

Cosine similarity between congruent and contradictory hidden states, same entity, vs. cosine similarity between two different entities’ congruent hidden states (the “does a lie perturb the representation more than an ordinary topic change” test), Qwen3-4B, layers 12/36 and 30/36:

| Layer | Same-entity (congruent vs. contradictory) | Cross-entity (congruent vs. congruent) |
|-------|-------------------------------------------|----------------------------------------|
| 12    | 0.9974                                    | 0.9969                                 |
| 30    | <b>0.9277</b>                             | <b>0.9829</b>                          |

Replicated across 3 relations  $\times$  3 seeds (mid-layer gap  $\approx 0$  on all 9 runs, late-layer gap positive on all 9), and on two further, architecturally different models: Phi-3-mini (layers 11/32 and 27/32) and Llama-3.2-3B-Instruct (layers 9/28 and 23/28) — same qualitative pattern on both, 9/9 relation-model combinations now agreeing on direction across all three model families:

| Model        | Relation   | Override % | Mid-layer gap | Late-layer gap |
|--------------|------------|------------|---------------|----------------|
| Qwen3-4B     | capital_of | 71%        | −0.0006       | <b>0.0525</b>  |
| Qwen3-4B     | founded_by | 98%        | −0.0035       | <b>0.0156</b>  |
| Qwen3-4B     | written_by | 84%        | −0.0048       | <b>0.0225</b>  |
| Phi-3-mini   | capital_of | 22%        | −0.0021       | <b>0.0214</b>  |
| Phi-3-mini   | founded_by | 51%        | −0.0067       | <b>0.0165</b>  |
| Phi-3-mini   | written_by | 64%        | −0.0091       | <b>0.0430</b>  |
| Llama-3.2-3B | capital_of | 89%        | 0.0032        | <b>0.0612</b>  |
| Llama-3.2-3B | founded_by | 16%        | −0.0074       | <b>0.0816</b>  |
| Llama-3.2-3B | written_by | 16%        | −0.0156       | <b>0.1681</b>  |

Relation-level magnitude is *not* stable across models: Qwen’s largest effect is **capital\_of** (0.053), Phi-3’s largest is **written\_by** (0.043), Llama-3’s largest is also **written\_by** but at 0.168 — nearly 3 $\times$  Phi-3’s — magnitude depends on the model, not solely the relation, and Llama-3’s effect sizes are uniformly the largest of the three models on this single-sentence framing.

### 2.3 Rigidity hypothesis — refuted at fact-level granularity

Proposed mechanism: facts held with a “deeper parametric attractor” should decohere more under override. Tested directly: per-fact teacher-forced confidence in the true tail (bare prompt, no document) against per-fact layer-30 decoherence,  $n = 45$  facts pooled across all 3 relations. Spearman  $\rho = 0.006$  — no relationship. Confidence vs. override-resistance:  $\rho = -0.225$ , correct direction but below the  $\sim 0.29$  threshold  $n = 45$  needs for significance. The relation-level magnitude pattern in §2.2 is real but not explained by per-fact memorization strength.

## 2.4 Active-lie hypothesis — refuted with real statistical power

Proposed alternative: decoherence tracks the model successfully *committing* to the false answer, not merely processing the contradiction. Tested on Phi-3 (lower compliance rate gives a naturally large stubborn/compliant split:  $n = 73$  stubborn,  $n = 62$  compliant, vs. only  $n = 5$  stubborn on Qwen):

| Layer | Stubborn (resisted) | Compliant (followed) | Baseline (no conflict) |
|-------|---------------------|----------------------|------------------------|
| 11    | 0.0036              | 0.0036               | 0.0095                 |
| 27    | <b>0.1018</b>       | <b>0.0747</b>        | <b>0.0623</b>          |

Stubborn > compliant > baseline at the late layer — the opposite of what an active-lie mechanism predicts. Resisting an explicit contextual instruction to preserve a parametric prior costs more geometric disruption than complying with it. At the mid layer, both groups score *below* the cross-entity baseline, which is a lexical artifact (swapping the whole subject perturbs shallow-layer tokens more than swapping only the claimed answer for the same subject) rather than evidence about conflict processing.

## 2.5 Reconciling the mid-layer null with Zhao et al. (2024)

Zhao et al. (arXiv:2410.16090) report  $\sim 90\%$  accuracy from a *trained* linear probe detecting conflict-presence at intermediate layers of Llama3-8B — in apparent tension with our untrained mid-layer null (§2.2). Replicating their actual method (trained logistic regression, congruent=0/contradictory=1, held out on unseen entities) on our 45-fact Qwen set: AUROC 0.642 at layer 12, 1.000 at layer 30. A trained probe *does* recover a real signal our untrained geometric check missed — the discrepancy is substantially a method difference — but 0.642 falls well short of their  $\sim 90\%$ , plausibly due to this replication’s much smaller dataset and a single train/held-out split with no cross-validation, not a clean quantitative match.

## 2.6 Realistic multi-sentence documents — the effect mostly does not survive

Every result above uses an isolated, single-sentence document: Document: France is a country whose capital is Munich. Real RAG documents are longer and contain other true content alongside the fact being queried. Scaled to 120 facts across 6 relations (capital\_of, founded\_by, written\_by, painted\_by, directed\_by, composed\_by, 20 facts each) with documents built as an opener + 2 *other true facts from the same relation* + the target sentence + a closer (congruent and contradictory versions share the identical filler facts, differing only in the target sentence’s tail — verified by construction, and a first version of this experiment had a bug letting filler facts differ between the two versions; fixed and rerun before trusting the result below), Qwen3-4B-Instruct:

| Relation    | Override % | Mid-layer gap | Late-layer gap |
|-------------|------------|---------------|----------------|
| capital_of  | 35%        | −0.0011       | <b>0.0048</b>  |
| founded_by  | 58%        | −0.0032       | −0.0105        |
| written_by  | 88%        | −0.0040       | −0.0177        |
| painted_by  | 92%        | −0.0039       | −0.0179        |
| directed_by | 85%        | −0.0033       | −0.0200        |
| composed_by | 95%        | −0.0050       | −0.0171        |

The late-layer gap survives for only 1 of 6 relations (`capital_of`, and even there it shrinks roughly 10 $\times$  from the single-sentence version’s 0.0525 to 0.0048) and *reverses sign* for the other 5. This is the central limitation of the whole result: the effect documented in §§2.2–2.4 is largely specific to isolated single-sentence contradictions and does not straightforwardly generalize to documents containing other true content alongside the target fact — the realistic case a deployed RAG verifier would actually need to handle. A plausible mechanism, not yet tested directly: once a document contains multiple true facts alongside one manipulated one, most of the document’s content is identical between the congruent and contradictory versions, which may dominate the late-layer representation and wash out or invert the effect of the one changed sentence relative to switching to a document about a completely different entity.

### 3 What is and isn’t shown

- On isolated single-sentence contradictions, a real, cross-model (3 architecture families), cross-relation late-layer geometric signal exists and is not explained by per-fact memorization strength or by whether the model ultimately complies.
- **That signal mostly does not survive realistic multi-sentence documents (§2.6)** — 1 of 6 relations keeps a (much smaller) positive gap, 5 reverse sign. This is the headline limitation, not a footnote: whatever this signal is, it is substantially a property of isolated single-fact framing, not yet a property of realistic documents.
- The magnitude of the single-sentence effect is model- and relation-dependent by a factor of up to  $\sim 3\times$ ; no mechanism tested so far explains that variation, and it is now moot for 5 of 6 relations once documents are realistic anyway.
- The practical framing that survives on single-sentence documents is **not** “lie detector” — it is closer to a **resistance/ friction detector**: it appears to fire more when a model fights its own parametric prior than when it quietly updates to match new context. Whether *any* framing survives on realistic documents is now the open question.
- Not shown: what explains the magnitude variation on single-sentence documents; what explains the sign reversal on realistic documents; whether either transfers past 6 relation templates and 3 model families; a deployed detection threshold (none is proposed here, and §2.6 makes one look premature).

### 4 Honest scope & limitations

- **The realistic-document result (§2.6) is the main limitation, not an afterthought.** The signal this note is built around is largely specific to isolated single-sentence contradictions. Anyone citing the §2.2–2.4 numbers should cite §2.6 in the same breath.
- **Moderate  $n$ .** 120 facts / 6 relations for the realistic-document result; 45 facts / 3 relations for the single-sentence cross-model result. Real effects at this scale, not yet large-sample claims.
- **Two models failed the override premise entirely.** GPT-2-base and TinyLlama-1.1B-Chat do not reliably follow in-context overrides at all and were excluded rather than forced; worth noting as a capability floor for any downstream use of this signal. A single worked

example is not sufficient evidence a model passes this premise either way — one Llama-3 example looked like a failure and was not representative of its 45-fact aggregate rate (§2.2).

- **No deployed detector.** This note establishes a measurable phenomenon (fragile even on the axis it was found on), not a calibrated, cross-domain threshold ready to gate real generations.
- **Prior art.** Zhao et al. (2024) already show conflict-presence is probeable at intermediate layers; this note’s distinct contributions are the late-layer resistance-vs-compliance split (§2.4), the negative result on per-fact rigidity (§2.3), and the negative result on realistic-document generalization (§2.6) — not the base claim that conflict leaves a residual-stream trace.

## 5 Reproduce

```
git clone https://github.com/HemanthKiranPolu/nous-decoherence
cd nous-decoherence
pip install -r requirements.txt
python -m nous.verifier_real_llm_test           # 2.1
python -m nous.verifier_in_context_conflict     # 2.2, Qwen
python -m nous.verifier_multi_relation_robustness # 2.2, 3-rel x 3-seed Qwen
python -m nous.rigidity_hypothesis_test         # 2.3
python -m nous.verifier_cross_model_phi3        # 2.2, Phi-3
python -m nous.verifier_cross_model_llama3      # 2.2, Llama-3 (HF_TOKEN)
python -m nous.verifier_compliance_split_phi3   # 2.4
python -m nous.replicate_zhao_conflict_probe    # 2.5
python -m nous.verifier_scaled_realistic        # 2.6, 120-fact realistic docs
```

## 6 What would make this result matter again

Section 2.6 is not a call for more relations or more seeds at the same design — it is a falsification of the practical version of the claim. Restoring it requires one of two things, not incremental replication of what is already here:

1. **A signal that survives realistic documents.** Not this cosine-similarity-on-raw-activations measurement at a fixed layer, evaluated on documents with other true content around the target fact — something that isolates the target fact’s contribution from the filler’s, rather than comparing whole-document hidden states where filler content dominates by volume.
2. **A mechanistic account of the collapse that generalizes.** “The identical filler facts probably dominate the representation” (§2.6) is a plausible guess, stated as one, not tested. A real account would predict *in advance* — from document length, filler/target ratio, or something else measurable — which relations keep a signal and which invert, rather than describing the six numbers already observed after the fact.

Neither is a small follow-up experiment. Absent one of these, the honest summary of this note is: a real, replicated, mechanistically-unexplained artifact of single-sentence framing, refuted twice on its proposed mechanism, and refuted once on its practical applicability. That is a legitimate negative result. It is not, on the evidence collected here, a verifier.

## References

- [1] Zhao et al., *Analysing the Residual Stream of Language Models Under Knowledge Conflicts*, arXiv:2410.16090 (2024).
- [2] Li et al., *SHIFT: Gate-Modulated Activation Steering for Knowledge Conflict Mitigation in Retrieval-Augmented Generation*, arXiv:2606.27786 (2026).